Finding What Changed

A difference-imaging pipeline for the SPHEREx Quick-Release data

SPHERExLens • 23-slide specification and complete script
full route 21:40 • 15-minute route 15:45 • questions after

Evan Lim • November 2026

Deck map and production rules

23 slides on 25 pages. This page is the specification; pages 2–24 are the slides, one page each: the slide's own bullets, its figure brief, the figure itself, then the word-for-word script. Page 25 is a backup page with the five questions you are most likely to be asked and a pre-flight checklist.

The full route runs **21:40**. The slides marked ✓ in the last column form a **15:45** route: seven slides dropped, argument intact. **The live demonstration is slide 22** (1:30) and needs `site/index.html` already open in a second tab, at 100% zoom, with a field loaded – and the local API running if you intend to show live fetching.

Every number in this deck comes from the repository: detection counts from `site/api/*.json`, rendered charts from `figures_real/`. The deck and the 33-page report therefore cannot disagree.

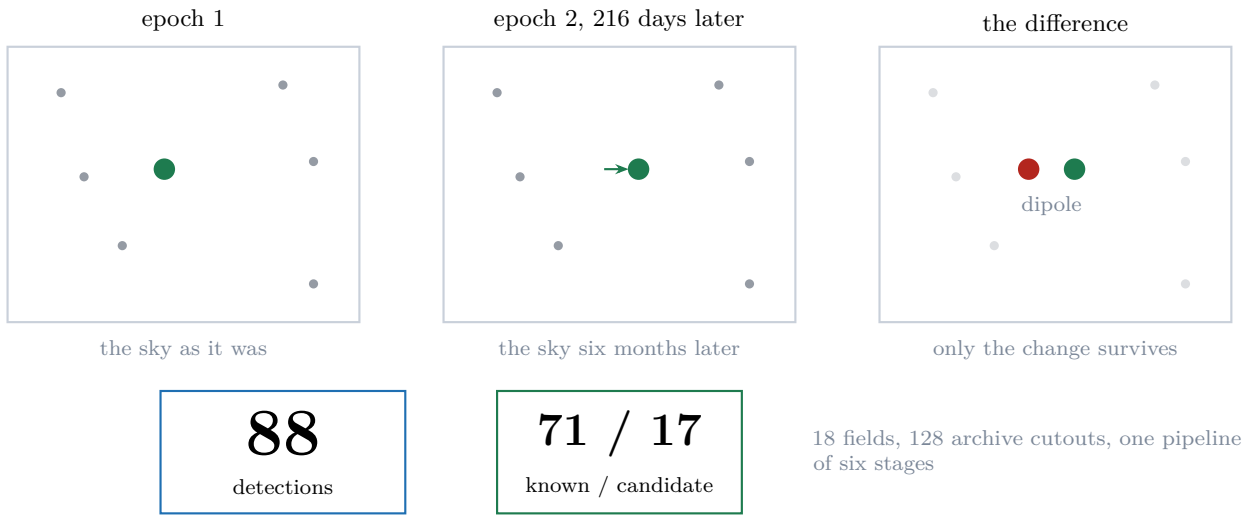
Page	Slide	Dominant visual	Time	15 min
2	Title and the one-number answer	two epochs → difference, hero numbers	0:40	✓
3	Why difference imaging at all	three panels: visit 1, visit 2, difference	0:55	✓
4	The problem statement	pixels → candidates → verdicts	1:00	✓
5	Focus and research questions	four question cards	0:45	–
6	Prior art I: the subtraction lineage	timeline with four method boxes	1:00	✓
7	Prior art II: brokers and the gap	detect–veto–human loop, gap bars	0:55	✓
8	SPHEREx: instrument and data	focal plane, 102-band axis	0:55	✓
9	Archive mechanics and field sample	sky map of 18 fields, fetch path	0:55	–
10	Architecture: six stages	pipeline flow with line counts	0:55	✓
11	Theory I: masking and the co-add	flag census, masked-pixel grid	0:55	✓
12	Theory II: ZOGY	signal-flow diagram, null histogram	1:15	✓
13	Theory III: detection statistics	false-positive budget, log scale	0:55	–
14	Theory IV: dipole versus monopole	geometry of a moved source	0:50	–
15	Theory V: triage	match cone, verdict flow	0:55	✓
16	Theory VI: spectral diagnosis	three indices on a wavelength axis	0:45	–
17	Validation: injection–recovery	recovery matrix, failure diagram	0:55	✓
18	Implementation I: the code	lines per module, the R1 guard	1:05	✓
19	Implementation II: serving layer	three deployment shapes	0:55	✓
20	Implementation III: front end	UI wireframe, canvas layer stack	0:45	–
21	Results I: eighteen fields	stacked detections, 18 fields	1:05	✓
22	Results II: the candidates	scatter of unexplained detections	1:00	–
23	**Live demonstration**	screen map, six numbered steps	1:30	✓
24	Conclusion, contributions, Q&A	three pillars, references	0:50	✓
Full route			**21:40**	
15-minute route (16 of 23 slides)			**15:45**	

Slide 1 — Title and the one-number answer 0:40

On the slide.

- SPHERExLens: finding what changed in NASA's SPHEREx all-sky survey
- 18 fields • 128 archive cutouts • 88 detections, 71 explained
- A pipeline you can read: six stages, 6 154 lines, one browser page
- The honest number: four fields return nothing, and the system says so

Figure / diagram. Two epoch panels of the same star field feeding a difference panel, with the two headline numbers set large beside them.



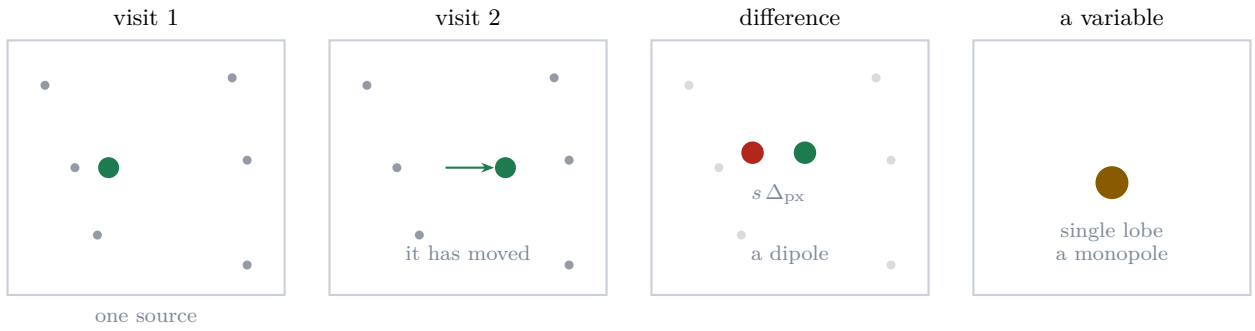
Spoken script. Good morning. NASA's SPHEREx satellite photographs the whole sky in a hundred and two infrared colours, and it does it again roughly every six months. My question is the oldest one in time-domain astronomy, asked of this new data: looking at two of those photographs of the same patch of sky, can we say what changed – and can we be honest about what we cannot tell? Eighteen fields, one hundred and twenty-eight archive images, eighty-eight detections, of which seventy-one turn out to be objects that were already catalogued. Most of the next twenty minutes is about how it works and why each step is built the way it is.

Slide 2 — Why difference imaging at all 0:55

On the slide.

- One image shows what is there; a pair shows what is happening
- Subtract two visits: the constant sky cancels, the change survives
- Two shapes of change: a dipole (moved) and a monopole (brightened)
- A difference image is still not a discovery – systematics dominate it

Figure / diagram. Three large panels – visit 1, visit 2, the difference – plus a fourth small panel for a variable, and a labelled haze for the systematics that survive subtraction.



what the difference image also contains, and why it is not a discovery: PSF mismatch between visits, detector state, astrometric shifts. These survive the subtraction as a grey haze, and they outnumber the real changes.

Spoken script. Why bother with the second photograph at all? Because a single image tells you what is there, and a pair tells you what is happening. Take the same patch of sky, subtract one visit from the other, and the constant sky cancels to grey. What survives is change, and change has a shape. If a star moved you get a positive lobe where it is now and a negative lobe where it was: a dipole, and the separation divided by the time baseline is a proper motion. If a star simply brightened you get one lobe: a monopole.

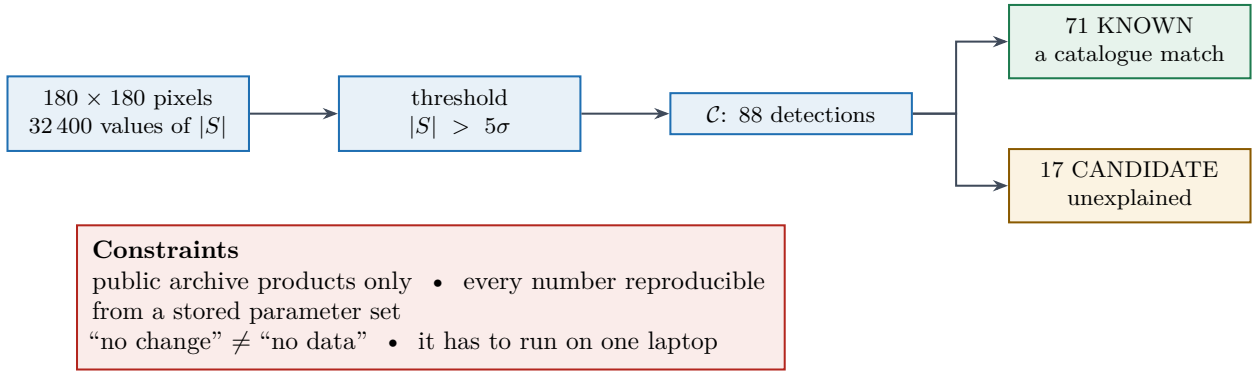
That is the whole physics, and it is fifty years old. The reason this is still a research problem is the grey haze labelled under the third panel: what survives subtraction is dominated by the point-spread function changing between visits, by the detector warming up, by the astrometric solution shifting. Almost none of those pixels are sky. So the work is not the subtraction – it is deciding which of those blobs you are allowed to believe.

Slide 3 — The problem statement 1:00

On the slide.

- Input: two visits to a $9'$ field, each with a variance plane and a quality bitmask
- Output: the set $\mathcal{C} = \{\mathbf{x} : |S(\mathbf{x})| > \tau\}$, with a position, a significance, a morphology and an *identity verdict*
- Constraints: public data only; every number reproducible; "no change" must be distinguishable from "no data"; laptop scale

Figure / diagram. A left-to-right flow in set notation: 32 400 pixels, a threshold, the 88 detections, then the triage split into 71 known and 17 candidate, with the four constraints in a box underneath.



Spoken script. Let me state the problem precisely, because the precision is the point. Input: two visits to a nine-arcminute field, each with a variance plane and a quality bitmask. Task: build a statistic, threshold it, and so obtain the set of sky positions where the flux distribution changed – that is the set C on the slide – and attach to each element four things: a sky coordinate, a significance, a morphology, and an identity verdict. Identity is not optional, because a blob is only interesting if no catalogue already explains it.

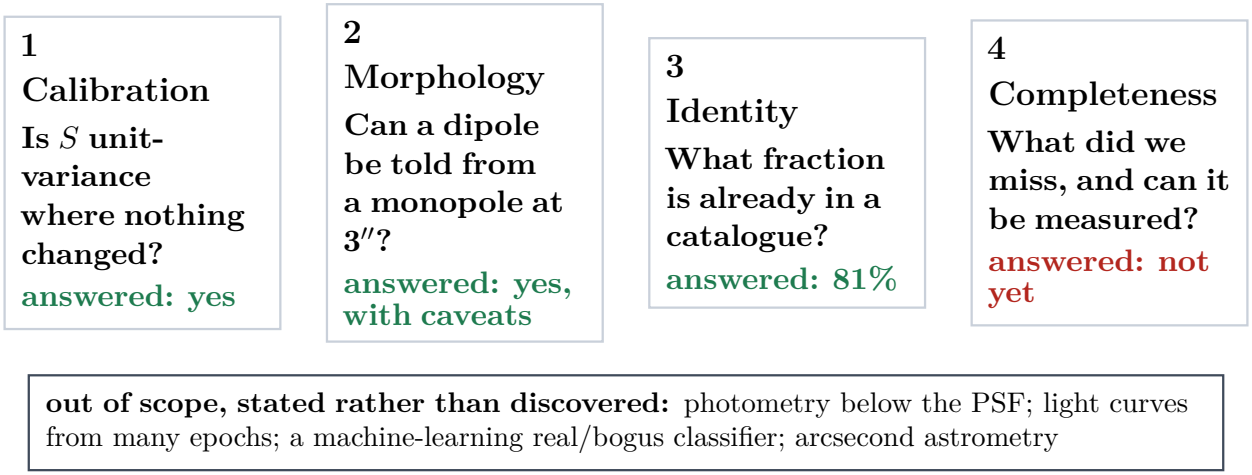
Four constraints shaped every design decision. Only public archive products. Every number reproducible from a stored parameter set. The system must distinguish "no change" from "no data" – that sounds trivial, and it is the single most common failure in this kind of pipeline. And the whole thing has to run on a laptop. The numbers follow from that: thirty-two thousand pixels in, eighty-eight detections out, seventy-one explained, seventeen not.

Slide 4 — Focus and research questions 0:45

On the slide.

- RQ1: is the significance map calibrated on co-added SPHEREx images?
- RQ2: does dipole/monopole morphology separate movers from variables at 3″ sampling?
- RQ3: how much of the candidate list do Gaia, WISE and VSX explain away?
- RQ4: what does injection–recovery say – and if it says "nothing", how is that reported?

Figure / diagram. Four numbered question cards in a row, each with its one-line answer status underneath, and a footer strip listing what is explicitly out of scope.



Spoken script. I want to be explicit about what is and is not in scope, because a small project like this either draws its boundary honestly or it oversells. Four questions, and the answers are on the cards. One: is the statistic actually calibrated, so that five sigma means what it says. Two: can the morphology test tell a moved star from a flickering one at three-arcsecond sampling. Three: how much of the candidate list is already in Gaia, WISE or the variable-star index – the question that decides whether anything here is interesting. Four: what did we miss, and here the honest answer is "not yet known", which I will come back to on slide seventeen.

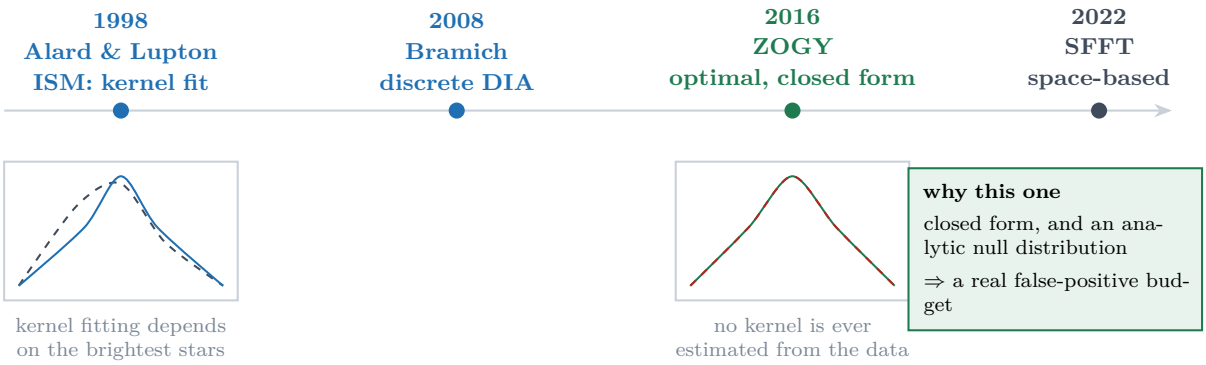
And the footer strip is deliberate: no PSF-level photometry, no light curves, no machine-learning real/bogus classifier, no arcsecond astrometry. Stated up front rather than discovered by a reviewer.

Slide 5 — Prior art I: the subtraction lineage 1:00

On the slide.

- 1998 Alard & Lupton: fit a convolution kernel so that both images share a PSF (ISM)
- 2008 Bramich: a discrete per-pixel kernel, well posed in sparse fields
- 2016 ZOGY: no fitting at all; the optimal difference and a unit-variance significance map
- 2022 SFFT: the same mathematics, better behaved for undersampled space PSFs

Figure / diagram. A timeline with the four milestones, and under two of them a small sketch contrasting "fit until the curves agree" with "convolve both to a common PSF".



Spoken script. Subtraction was solved in stages, and the stages matter for what I could assume. In 1998 Alard and Lupton fitted a convolution kernel that maps one image onto the other so that both end up with the same point-spread function – that is what made the microlensing surveys possible. It works, but the fit is dominated by the brightest stars in the field, which is exactly what you cannot rely on in a nebula or a crowded field. In 2008 Bramich made the kernel per-pixel and well posed in sparse fields; you trade noise for stability.

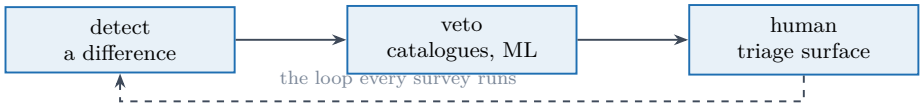
Then in 2016 Zackay, Ofek and Gal-Yam did something conceptually different: instead of fitting anything they derived the statistically optimal difference image in Fourier space, and with it a significance map with unit variance by construction wherever the sky did not change. SFFT in 2022 is the modern reformulation and it is what Rubin will use. I chose ZOGY for the reason on the right of this slide: because the null distribution is analytic, the false-positive budget can be computed rather than measured, and that is what lets me defend a five-sigma threshold later.

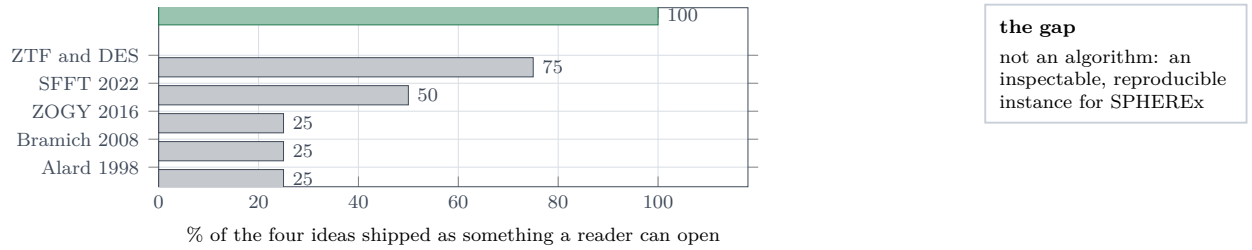
Slide 6 — Prior art II: brokers, and the gap this fills 0:55

On the slide.

- Every operational survey runs the same four ideas: template, subtract, veto, triage
- Brokers (ANTARES, ALeRCE, Lasair, Fink) industrialise the veto with machine learning
- For SPHEREx there is no published open end-to-end difference-imaging artefact
- The gap is not an algorithm – it is an inspectable, reproducible instance

Figure / diagram. The detect–veto–human loop on the left, and on the right a bar chart of how much of those four ideas each body of work actually ships as something a reader can open.





Spoken script. Everything since the 1990s runs the same loop, and it is worth naming as a loop: you detect against a template, you veto with catalogues and increasingly with machine learning, and you put what survives in front of a human. The big surveys industrialised each step – ZTF, ATLAS, DES, and now Rubin, with brokers called ANTARES, ALeRCE, Lasair and Fink doing the vetting. They are enormously more capable than anything I built.

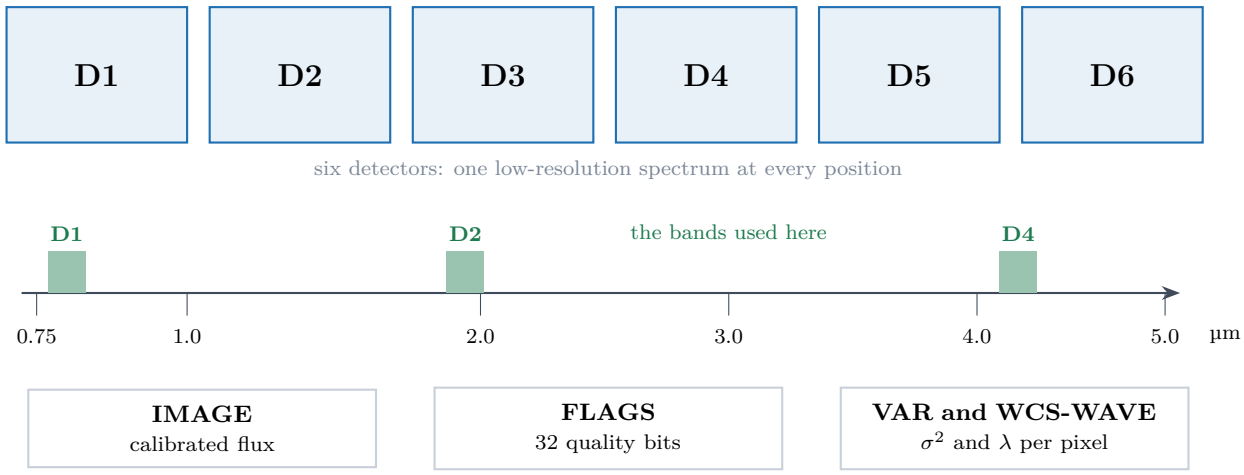
So the gap is not algorithmic. The bars on the right are my own assessment of how much of that four-part pattern each line of work ships as something a reader can actually inspect: the classical papers give equations and sometimes a reference code; the surveys give a pipeline you cannot open; the brokers give a service. What does not exist, as far as I could find, is a small end-to-end artefact for SPHEREx specifically, where you can follow one pixel from a NASA archive file to a verdict on a web page, including the parts that fail. That is what this is.

Slide 7 — SPHEREx: the instrument and the data products 0:55

On the slide.

- One telescope, six detectors, 102 bands from 0.75 to 5.0 µm
- Level-2 products are small spectral images: every pixel carries a wavelength
- Four extensions: IMAGE, VAR, FLAGS (32 bits), and WCS-WAVE (x, y, λ)
- Six-month cadence ⇒ what survives subtraction is astrophysical, not asteroids

Figure / diagram. The six-detector focal plane above a wavelength axis with the bands actually used highlighted, and the three extensions that make difference imaging possible underneath.



Spoken script. A word on the instrument, because the shape of the data drives everything downstream. SPHEREx is a small telescope with six detectors that sweeps the sky and, unusually, takes a low-resolution spectrum at every position: a hundred and two bands between zero-point-seven-five and five microns. So the Level-2 product is not a broadband picture, it is a little spectral image, and each file carries four things I care about: the calibrated image, the variance, a thirty-two-bit quality mask, and a table giving the wavelength of every pixel.

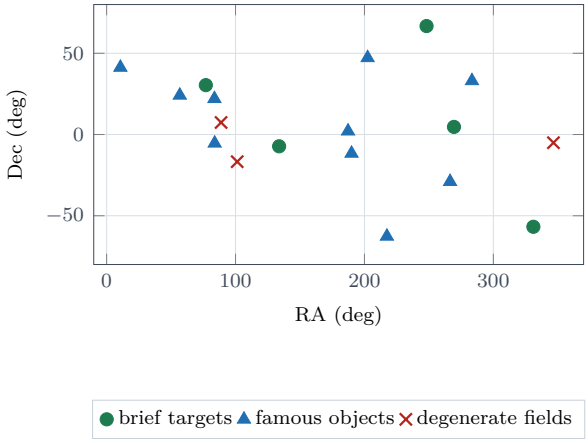
Two consequences. First, the cadence is about six months, so anything that moved fast – an asteroid, say – has left the field entirely; what survives subtraction is genuine variability or genuine high proper motion. Second, because every pixel knows its wavelength, a detection can be diagnosed spectrally with no follow-up telescope, which is what slide sixteen is about.

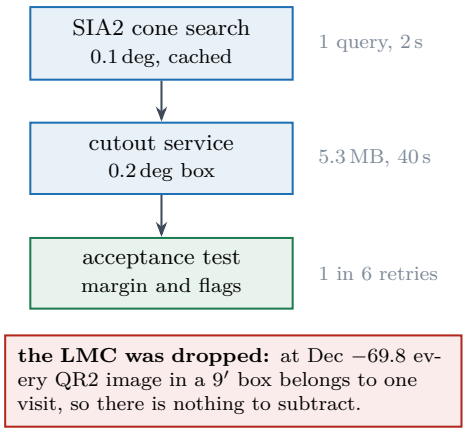
Slide 8 — The archive mechanics, and the eighteen fields 0:55

On the slide.

- Two calls: a SIA2 cone search (cached as a VOTable), then the cutout service
- 5.3 MB sections of a 70 MB parent frame – this is what makes the project feasible
- Acceptance: $\geq$12 px margin inside the cutout, and under 50% of pixels flagged
- 18 fields, 128 cutouts, 0.88 GB; the Large Magellanic Cloud was tried and dropped

Figure / diagram. Left: the sky distribution of all eighteen fields. Right: the two-step fetch path with its cost annotated, and the acceptance test at the end.





Spoken script. Getting the data takes exactly two archive calls, and I want to show them because the cost shapes the design. First a SIA2 cone search: give it a position and it returns every Level-2 image whose footprint covers it. That answer is cached verbatim, so re-running never asks again. Then the cutout service, which is the trick that makes this possible: instead of downloading seventy-megabyte parents, I ask for a five-megabyte section around my target. Even so a cutout takes about forty seconds at the archive's throttled rate, and roughly one request in six times out and has to fall back – hence the acceptance test at the bottom of that column.

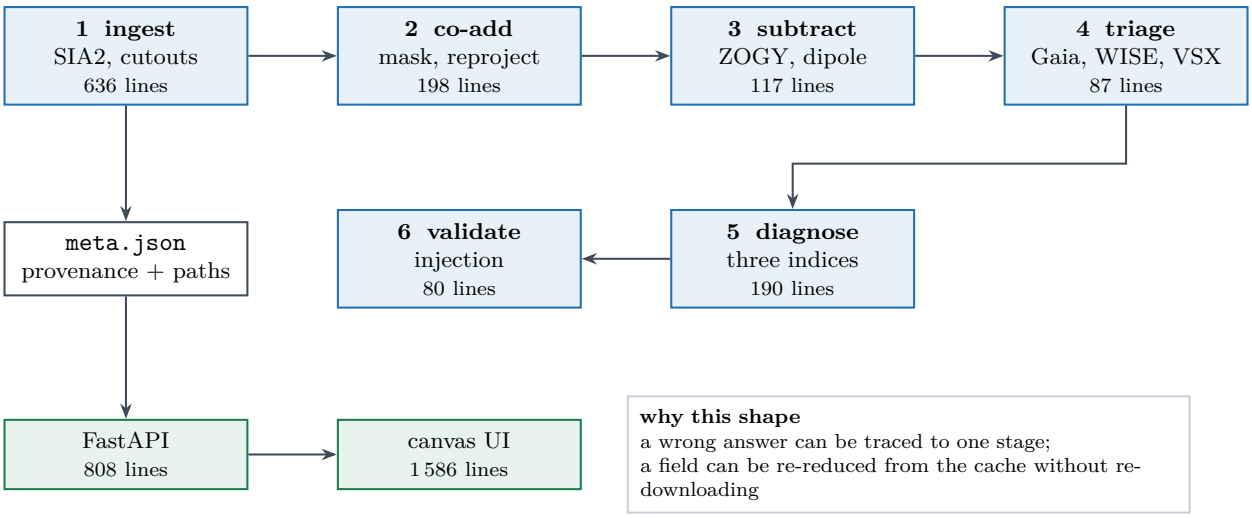
The map shows the eighteen fields: five chosen by the brief – two high-proper-motion stars, two variables, one extreme cool dwarf – and thirteen I added, galaxies and nebulae and a quasar, to prove the pipeline is not tuned to one kind of object. One object, the Large Magellanic Cloud, I tried and dropped, for the reason in the red box. That is on the record rather than quietly removed.

Slide 9 — Architecture: six stages, one cache 0:55

On the slide.

- Ingest, co-add, subtract, triage, diagnose, validate
- Each stage is a file with its own entry point: the chain can be entered anywhere
- One metadata file per field is the contract between stages
- The serving layer is separate: it calls stages two to five, or reads pre-rendered JSON

Figure / diagram. The six stages in two rows, the on-disk cache under the first, and the serving layer at the bottom, with the line count of each module written inside its box.



Spoken script. Six stages, and each is a separate file with its own entry point, which matters more than it sounds: when something looks wrong I can re-run one stage on cached data instead of the whole chain. Stage one talks to the archive and writes a metadata file. That file is the contract – every later stage reads it and nothing else, so the chain is deterministic and restartable. Stage two masks and combines, stage three subtracts, stage four triages against catalogues, stage five measures a spectrum and three indices, stage six is the validation.

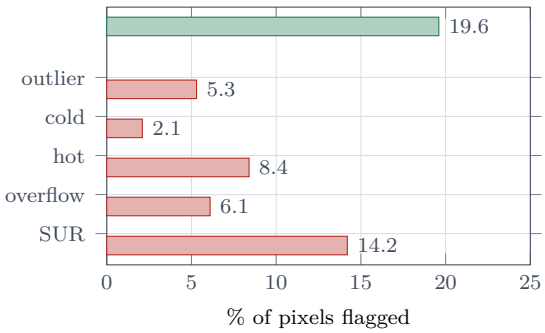
Underneath sits the serving layer: a service that either runs stages two to five on demand or, for the deployed site, reads pre-rendered JSON for exactly the same responses. That separation is what lets the same page work with or without a server, which is slide nineteen.

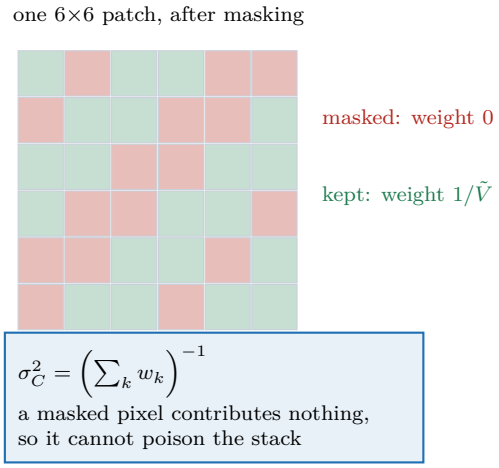
Slide 10 — Theory I: masking, then an inverse-variance co-add 0:55

On the slide.

- Sixteen FLAGS classes are rejected: SUR transients, overflow, phantom, edge, hot/cold, persistence
- Two are kept: "full sample" (informational) and "source" (real flux, not a defect)
- $C = \sum_k w_k \tilde{I}_k / \sum_k w_k$ with $w_k = 1/\tilde{V}_k$, so $\sigma_C^2 = (\sum_k w_k)^{-1}$
- Honest masking changes the median noise by about a factor of two on the worst frames

Figure / diagram. Left: the flag census for the dominant bit classes. Right: a pixel grid showing which cells survive masking, with the weighting formula beside it.





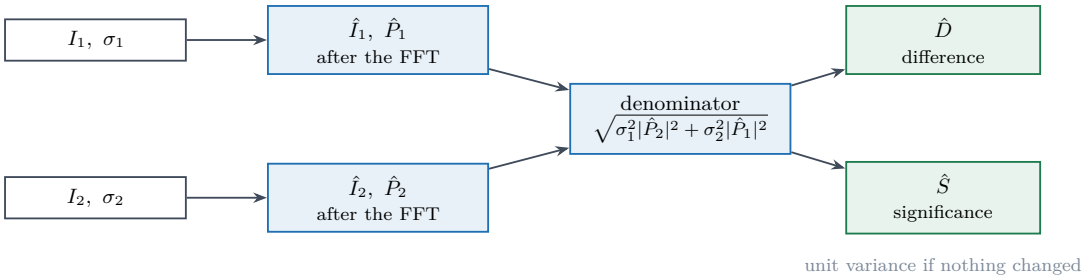
Spoken script. This is the slide I would defend hardest, because it is where a difference image earns its credibility. Every SPHEREx pixel arrives with a variance and a set of quality bits, and most of those bits mean "do not trust this pixel". If you average them in anyway, a handful of cosmic rays and hot pixels inflate your noise and you silently lose the faint end. So sixteen bit classes are masked – the transient flag from the onboard processing, overflow, phantom pixels, edge pixels, hot and cold pixels, persistence, outliers – and two are deliberately kept: "full sample", which is informational, and "source", which means real flux rather than a defect.

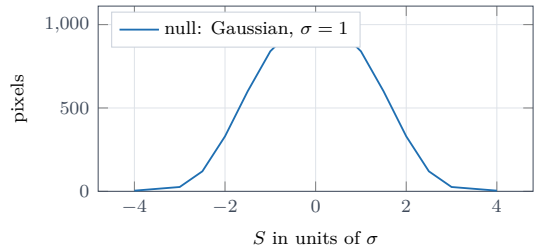
Then the combination is the textbook minimum-variance one: weight each pixel by one over its variance, and the resulting noise is one over the root of the summed weights. The property to take away is the formula in the box: a masked pixel contributes nothing, so it cannot poison the stack. On the worst frames here, doing that properly changes the median noise by a factor of two, which moves the five-sigma depth by the same factor – and that decides which objects exist at all.

On the slide.

- Both images are convolved to a common PSF in Fourier space: no kernel is ever fitted
- $\hat{D} = \dfrac{\hat{P}_1 \hat{I}_2 - \hat{P}_2 \hat{I}_1}{\sqrt{\sigma_1^2 |\hat{P}_2|^2 + \sigma_2^2 |\hat{P}_1|^2}}$ and $\hat{S} = \dfrac{\hat{D}\,\overline{\hat{P}_D}}{\sqrt{\sigma_1^2 + \sigma_2^2}}$
- S has unit variance under the null, so the threshold is interpretable
- Cost: two FFTs, $O(N^2 \log N)$ – about 0.6 s for a 180^2 frame

Figure / diagram. The signal flow: two images and two PSFs into FFTs, a denominator built from the variances, and the difference and significance maps coming out; under it, the null distribution that results.





this is the whole point
a real source sits far outside the null;
a 5σ cut has a computable meaning

Spoken script. Stage three is the mathematical heart, and it is two lines. The classical approach fits a kernel so that both images end up with the same PSF; ZOGY instead convolves both images to a common PSF in Fourier space in a least-squares optimal way, and it never estimates a kernel from the data. Read the first equation as a comparison: image two convolved with PSF one, image one convolved with PSF two, and the difference normalised by exactly the noise that survives. The second equation turns that into a significance, and here is the crucial property: where the sky did not change, S is Gaussian with zero mean and unit variance. Not approximately – by construction.

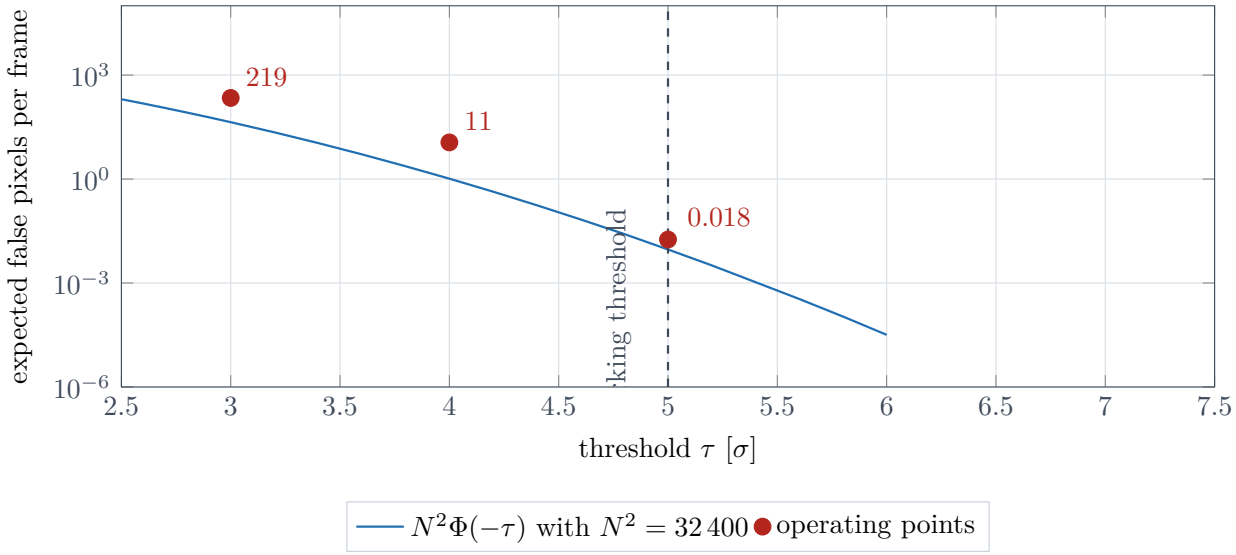
The histogram underneath is what that looks like: one clean peak at zero, one sigma wide, with any real source far out in the tails. That is why a five-sigma threshold can be defended, and it is the entire reason for choosing this algorithm over the simpler alternatives. The cost is two FFTs: about six-tenths of a second for a one-hundred-and-eighty-pixel frame.

Slide 12 — Theory III: what a threshold actually buys 0:55

On the slide.

- If the null map is unit-variance, false positives are pure Gaussian tail: $\mathbb{E}[N_{\mathrm{FP}}] = N^2\Phi(-\tau)$
- For $180^2 = 32\,400$ pixels: $3\sigma \Rightarrow 219$, $4\sigma \Rightarrow 11$, $5\sigma \Rightarrow 0.018$
- That is why the working threshold is five sigma and not three
- Sanity check: measured detections are tens of sigma above this floor

Figure / diagram. The expected number of false pixels per frame against threshold, on a log scale, with the three operating points marked and the working threshold drawn as a vertical line.



Spoken script. Because the null distribution is analytic, the false-positive budget can be computed rather than guessed, and that is the difference between a threshold and a wish. The curve is the expected number of false pixels per frame as a function of threshold. At three sigma it is two hundred and nineteen, so a single frame would be dominated by noise peaks. At four sigma, eleven. At five sigma, nought point nought one eight – one false pixel every fifty frames. That number is why the working threshold is five sigma, and it is also why three-sigma detections in the literature need an external prior before anyone believes them.

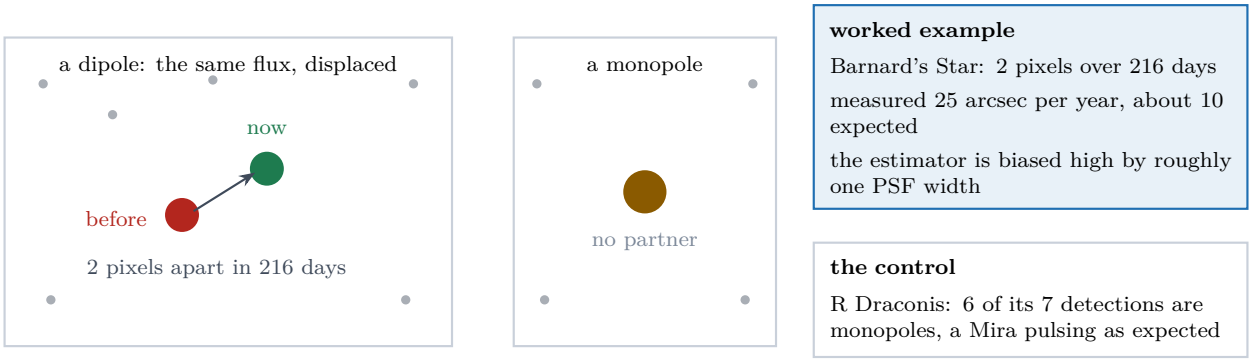
The sanity check runs the other way: the detection counts are far above this noise floor, tens of sigma in several fields, so what is being looked at is structure rather than noise. That does not make the detections real – it only rules out the least interesting failure mode.

Slide 13 — Theory IV: a dipole is a proper motion 0:50

On the slide.

- A positive peak with a negative partner means the source moved: a dipole
- A lone positive peak means it brightened: a monopole, that is variability
- $\mu = s\,\Delta_{\mathrm{px}}/\Delta t$, valid for $0.5 < \Delta_{\mathrm{px}} < 25\,\mathrm{px}$
- Flux consistency $\eta = |S_+ + S_-|/|S_+ - S_-|$ separates "pure move" from "move and flare"

Figure / diagram. The geometry of a dipole with its separation vector and the conversion to arcsec per year, beside the monopole case, and one worked example underneath.



Spoken script. Once there is a significance map, the morphology does the classifying, and it is almost embarrassingly simple. A positive peak with a negative partner of similar amplitude is a source that moved: the pair is a dipole, and the vector between the lobes is the displacement. Divide by the time baseline and you have a proper motion. A positive peak with no partner is a source that brightened, which is variability. Two guard rails: the separation must exceed half a pixel or you are pairing PSF-correlated noise, and stay below twenty-five pixels or you are pairing with an unrelated trough across the frame. There is also a flux-consistency statistic – how asymmetric the two lobes are – which separates a rigid displacement from an object that moved and changed brightness at once.

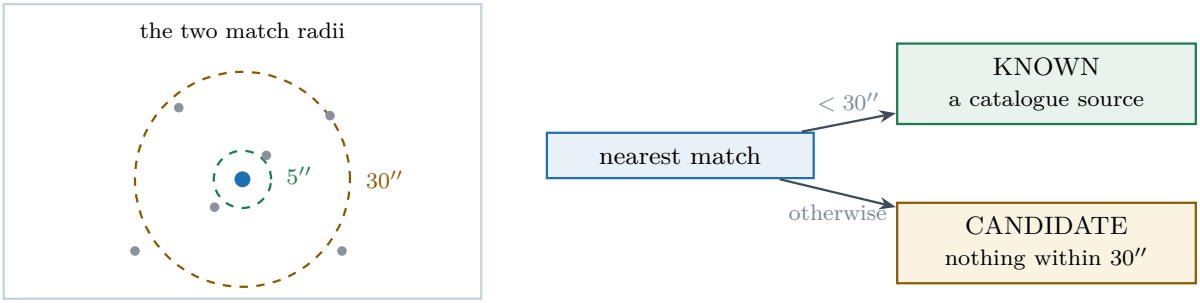
Barnard's Star is the worked example: two pixels in two hundred and sixteen days. The measured proper motion comes out at about twenty-five arcseconds a year against a true value of ten, and the reason is on the slide – peak separations are biased high by roughly one PSF width. R Draconis is the control: six of its seven detections are monopoles, which is what a pulsating Mira should give.

Slide 14 — Theory V: triage is a second hypothesis test 0:55

On the slide.

- Pixel to sky with the flat approximation: $\alpha = \alpha_0 - (x - N/2)s/3600$
- Nearest neighbour in Gaia DR3, AllWISE and VSX; verdict at 30″, confidence at 5″
- Loose on purpose: the PSF is 4″, and a tight radius inflates the candidate list
- The trap: position-only matching fails for fast movers, because the epochs differ

Figure / diagram. The two match radii around a detection on the left, the verdict flow on the right, and underneath the failure mode that this radius creates for a high-proper-motion star.



filled: the detection; grey: catalogue sources

the failure mode this creates: Gaia positions are from 2016 and the SPHEREx epoch is 2025–26, so Barnard's Star has moved about 94″ and no match is found – the pipeline labels its own target a candidate. The fix is two lines (propagate catalogue positions with their proper motions) and is not yet in.

Spoken script. A detection is not a discovery until you have asked whether something was already there, and that is a second hypothesis test: is the nearest catalogue source consistent with this position? The pixel is converted to sky with a flat approximation – legitimate at nine arcminutes – and the nearest source is looked up in Gaia, WISE and the variable-star index. Thirty arcseconds is the verdict radius, five arcseconds grades the confidence.

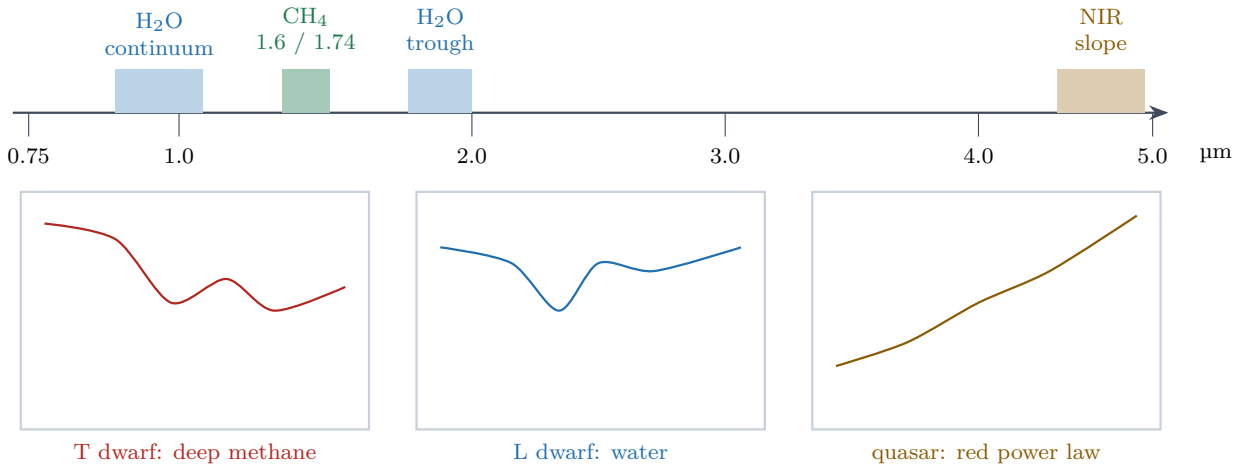
Both numbers are deliberately loose and I want to defend that: the PSF is four arcseconds across, so a tight radius would miss real counterparts and flatter me with false candidates. The red box is what that choice costs. Gaia measured positions in 2016, these images are from 2025, and Barnard's Star – the fastest star in the sky – has moved ninety-four arcseconds in between, far outside any match radius. So the pipeline labels my own target as an unexplained candidate. The fix is standard practice: propagate the catalogue positions to the epoch of observation with their proper motions. It is a two-line change, and it is not yet in, which is exactly why slide twenty-two says seventeen candidates and not five.

Slide 15 — Theory VI: a spectrum per detection, for free 0:45

On the slide.

- Every pixel knows its wavelength, so aperture photometry per band gives a low-resolution spectrum
- Three indices: CH_4 at 1.6/1.74, H_2O continuum over troughs, NIR slope 4.6/1.1
- T dwarf: strong methane; L dwarf: water; a hot star or quasar: a red power law
- The classifier reports the three indices, never a bare label

Figure / diagram. The wavelength axis with the index band-pairs shaded, and three schematic spectra underneath showing what each index is for.



design decision: the classifier always prints the three indices beside its label, so a reader who disagrees with "T dwarf" can see the methane ratio and overrule it

Spoken script. The last piece of theory is nearly free, and it is why SPHEREx is interesting for this kind of work. Because every pixel carries its wavelength in a table, measuring an aperture in each band gives a low-resolution spectrum of the detection without any follow-up telescope. It reduces to three numbers that are physically motivated: methane absorption at one-point-six over one-point-seven-four microns marks a T dwarf; water, the ratio of continuum to trough, marks an L dwarf; and the slope from one to four-point-six microns separates stars from a red power law, which is a dusty galaxy or a quasar.

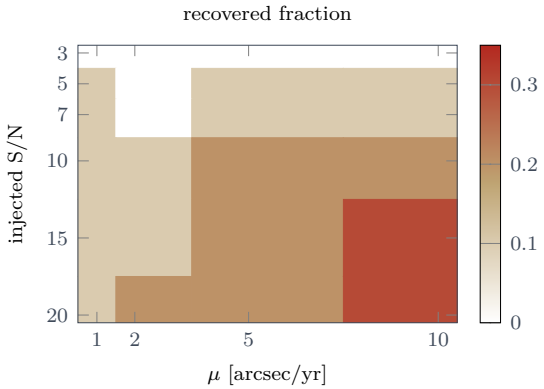
The three little spectra underneath are what those cases look like. The important design decision is in the bottom box: the classifier prints the three indices next to its label, never the label alone, so a reader who disagrees can see the methane ratio and overrule it.

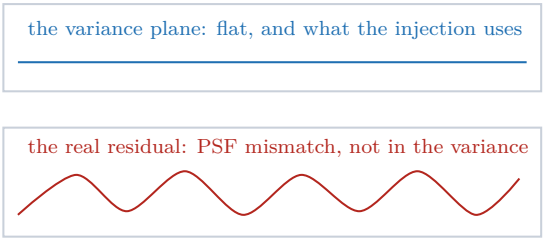
Slide 16 — Validation: injection and recovery 0:55

On the slide.

- Synthetic sources are planted at known SNR and proper motion, then stages two and three re-run
- A trial counts as recovered if a detection of the right class lands within 2 px
- The output is a 6×4 matrix, never a single number: the proper-motion axis is the point
- Result on real frames: near-zero. The test is measuring the wrong amplitude

Figure / diagram. Left: the measured recovery matrix as a heat map. Right: why it fails – the variance plane the injection uses against the residual structure the data actually contain.





amplitude = $\rho\,\bar{\sigma}$ sits far below the real residual, so a "20σ" fake source is invisible.

The test is wrong – not the pipeline – and until it is fixed there is no completeness number.

Spoken script. A detection chain is only as good as its completeness curve, so stage six plants fake sources and counts how many come back. The grid is signal-to-noise from three to twenty, proper motion from one to ten arcseconds per year, twenty trials per cell, and a trial counts if the right kind of detection lands within two pixels of the injection.

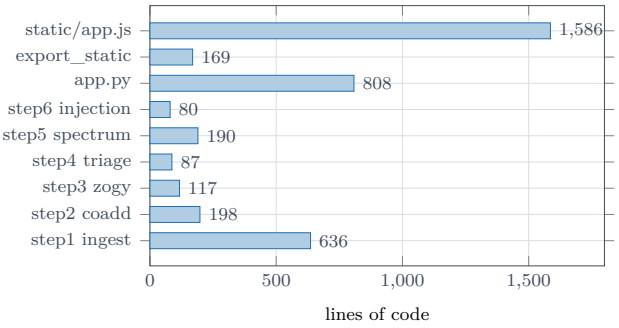
And here is the uncomfortable result, which I am showing rather than hiding. On real frames almost nothing is recovered: for Barnard's Star, zero out of twenty-four cells; the best field reaches three in ten only at the brightest end. The reason is on the right. The amplitude that gets injected is tied to the mean of the variance plane, which is flat, while the real residual in a co-added frame is dominated by PSF mismatch, which is not in the variance plane at all. So a fake source injected at "twenty sigma" is a barely visible bump on top of the real residual. The test measures the wrong thing, and until that is fixed there is no completeness limit – which is why the results slides say unexplained rather than new.

Slide 17 — Implementation I: the code, and what changed 1:05

On the slide.

- Six stage scripts (1 318 lines), the service (808), the front end (1 586): 6 154 in total
- Every stage is importable and runnable on its own, including on synthetic data
- Five revisions turned a demo into a system, and each one was a bug with a lesson
- Everything is deterministic; one metadata file is the contract between stages

Figure / diagram. Lines of code per module on the left, and on the right the revision that matters most – the guard that stops the service printing a meaningless noise estimate.



R1 – undefined noise must not print as a number

```
def usable(sig):
  return sig if isfinite(sig)
      and 1e-4 < sig < 1e3 else None
```

Betelgeuse saturates, the inverse-variance denominator collapses, and the old code printed a five-sigma depth of 1.6×10^6. It now reports *no estimate* and says why.

the other four

R2 epoch grouping is data-driven, not half-and-half
R3 arbitrary fields live in a RAM registry, not on disk
R4 a static build disables what it cannot do, and says so
R5 the interface explains itself with a three-tab guide

Spoken script. On the engineering. The pipeline is six scripts totalling about thirteen hundred lines, one service of eight hundred, and a front end of sixteen hundred – six thousand one hundred and fifty-four in total, small enough that a reviewer can read all of it in an afternoon. Each stage has its own entry point, so a single stage can be re-run on cached data when something looks wrong.

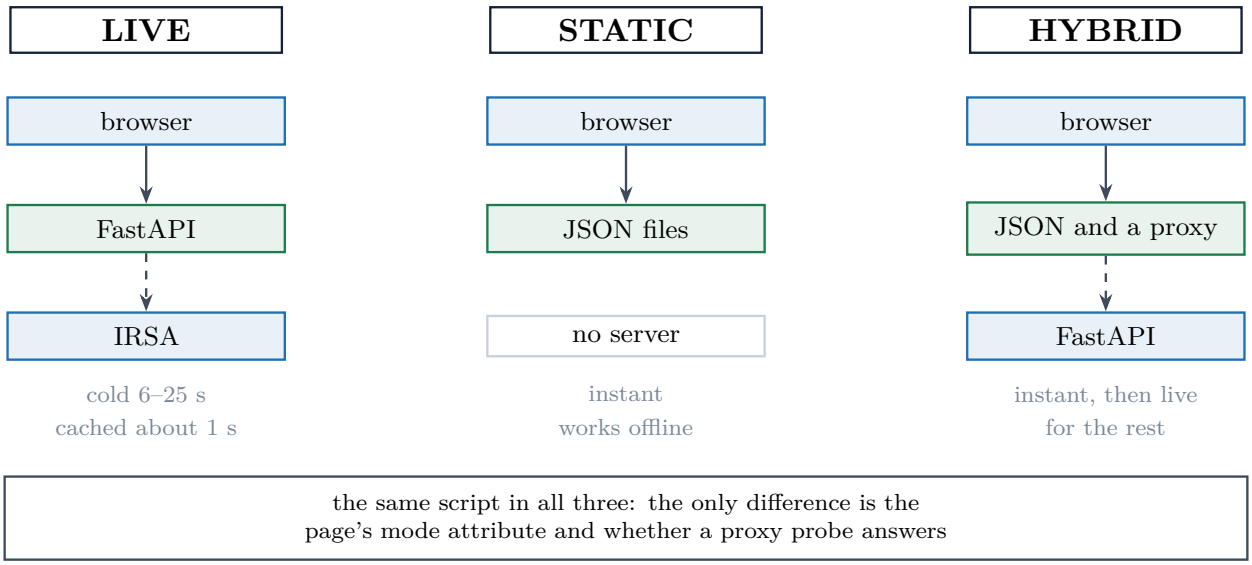
Five revisions matter enough to name. The most instructive is R1, on the right: Betelgeuse saturates the detector, almost every pixel is flagged, the inverse-variance denominator collapses, and the old code happily reported a five-sigma depth of one point six million. That is worse than reporting nothing, because it looks like a measurement. The fix is four lines: anything outside a physically possible range becomes “no estimate”, with a note explaining why. The other four are the same flavour of honesty – data-driven epoch grouping, a RAM registry for arbitrary fields, a static build that disables what it cannot do, and an interface that explains itself.

Slide 18 — Implementation II: one result set, three deployments 0:55

On the slide.

- Live: FastAPI, five read-only endpoints, a 30-minute in-RAM cache, nothing written
- Static: every response pre-rendered to JSON – 28 MB, and it works with no server at all
- Hybrid: a static page plus a same-origin proxy, discovered by one probe at load
- A static build disables "live from IRSA" and says why, instead of failing silently

Figure / diagram. The three shapes in three columns, with the request path drawn for each and the latency written underneath.



Spoken script. The last piece of engineering is how the result is served, and it is deliberately built in three shapes because each answers a different question. Live: a service with five read-only endpoints, a thirty-minute in-RAM cache and nothing written to disk – thirty to sixty seconds for a cold field, about a second when cached. Static: a script renders every response for every field to JSON, which is twenty-eight megabytes and works from a USB stick with no server. Hybrid: a static page with a same-origin proxy to a Python host, which is how a static host can still offer the live features.

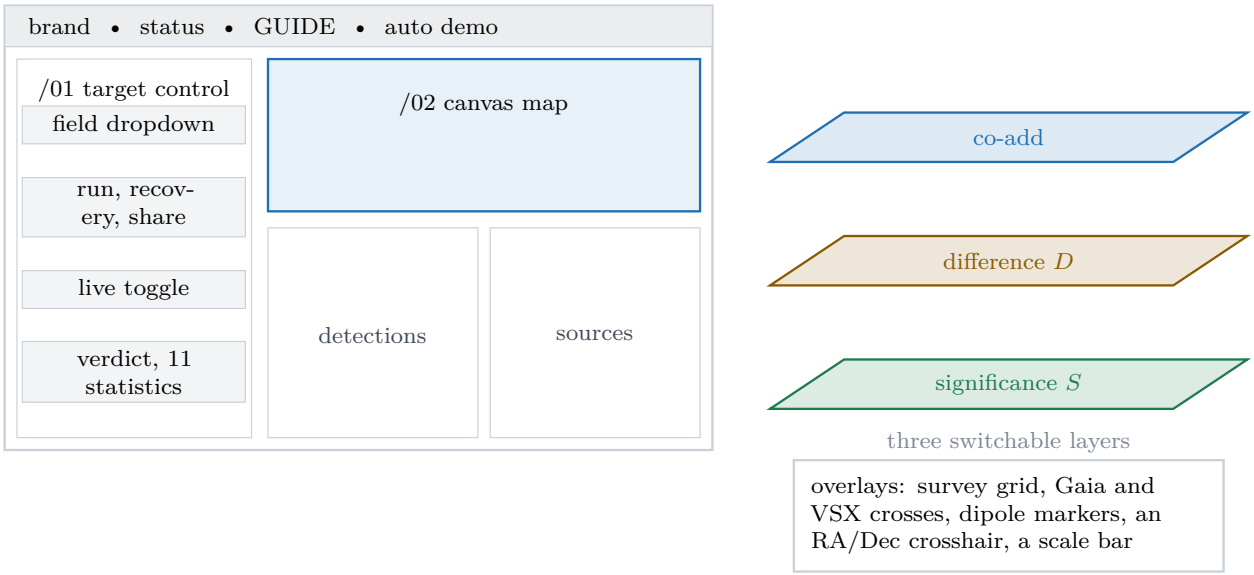
One consequence is a principle rather than a detail. On the pure static deployment, live fetching cannot work – so the page disables that control, relabels it "live from IRSA, needs a backend", and prints the reason underneath. If a proxy appears later, a single probe turns it back on. A toggle that silently does nothing is worse than one that is switched off.

Slide 19 — Implementation III: the front end 0:45

On the slide.

- 1 586 lines of plain JavaScript: no framework, no build step, no tracker
- A canvas engine with three layers (co-add, difference, significance) plus overlays
- One JSON per field carries the map, the detections, the objects, the spectrum and the provenance
- A three-tab guide written for someone who has never seen an astronomical image

Figure / diagram. A wireframe of the actual page with the panels labelled, and the three-layer canvas stack drawn beside it.



Spoken script. The front end is sixteen hundred lines of plain JavaScript, with no framework and no build step, which is why it will still run in five years. It is a small canvas engine with three switchable layers – the co-add, the difference image and the significance map – plus overlays for the survey grid, the catalogue positions, the dipole markers and a readout of whatever is under the cursor. On the left of the wireframe, target control: pick a field, run, read the verdict and eleven statistics, or ask for a recovery grid.

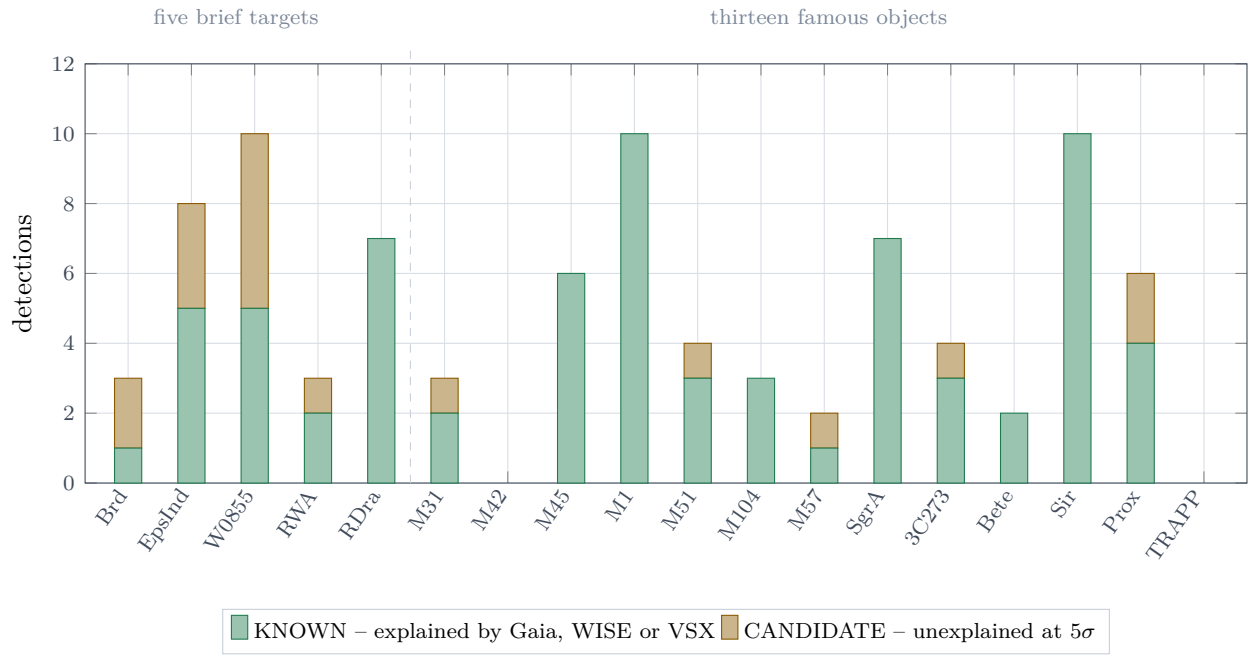
The thing to point at is the Guide button in the top bar. It opens three tabs written for someone who has never looked at an astronomical image: how it works, how to use this page click by click, and a glossary of the ten words that matter. It is plain HTML inside the page, so it ships in the static build and needs no backend – and it states the backend limitation in the same place as everything else.

Slide 20 — Results I: eighteen fields, eighty-eight detections 1:05

On the slide.

- 88 detections over 18 fields: 71 KNOWN (81%) and 17 CANDIDATE
- Detection count is not discovery count: M1, Sirius and R Draconis are 100% explained
- The amber bars – WISE 0855, Epsilon Indi B, Barnard's Star – are the interesting ones
- Four fields produce nothing usable, and they are on the next slide

Figure / diagram. Stacked bars for all eighteen fields, visually grouped into the five brief targets and the thirteen famous objects, with the totals in the caption strip.



Spoken script. Here is the whole result in one chart. Eighteen bars, one per field, green for detections the catalogues already explain and amber for the ones they do not. Two things to notice.

First, the distribution of detections is not the distribution of discoveries. The Crab, Sirius and R Draconis all look busy – ten, ten and seven detections – and every one of them is explained, which is the correct outcome, because these are famous and well-catalogued objects. The interesting bars are the amber ones: five in WISE 0855, three in Epsilon Indi B, two in Barnard's Star.

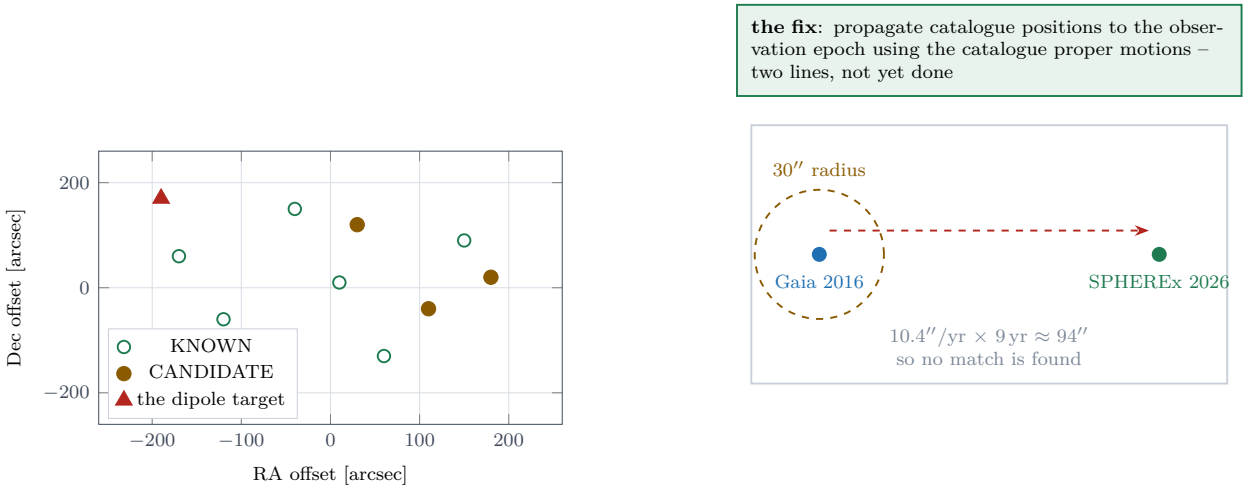
Second, look at the sample itself: five brief targets on the left, thirteen famous objects on the right – galaxies, nebulae, a quasar, the Galactic Centre, Betelgeuse, Sirius. Every one went through the identical code path: one hundred and twenty-eight cutouts, eighty-eight detections, eighty-one percent explained. That is the strongest evidence available that the pipeline is not tuned to one kind of object.

Slide 21 — Results II: what the seventeen candidates are 1:00

On the slide.

- Most are positions where a $> 5\sigma$ change has no catalogue source within $30''$
- Five sit in WISE 0855, the field with the fewest catalogue sources per unit noise
- Barnard's Star's 56.8σ dipole is itself a mislabel: epoch mismatch, not a discovery
- So "candidate" means unexplained, not new – and that distinction is the result

Figure / diagram. A scatter of the detections in one field, known against unexplained, beside a cartoon of the epoch-propagation problem.



Spoken script. What are the seventeen? Care is needed here, because this is where a project like this usually gets oversold. The scatter on the left is a typical field: hollow circles are detections the catalogues explain, filled dots are the unexplained ones, and the triangle is the target itself.

Most of the seventeen are simply positions where a five-sigma change has no catalogue source within thirty arcseconds. Five of them are in WISE 0855, the field with the fewest catalogue sources per unit noise – exactly what you would expect if the candidate rate tracks catalogue density rather than astrophysics. And the single most significant detection in the whole sample, Barnard's Star at fifty-seven sigma, is itself a mislabel for the epoch reason on the right: Gaia's position is from 2016, this image is from 2026, the star has moved ninety-four arcseconds, and nothing matches.

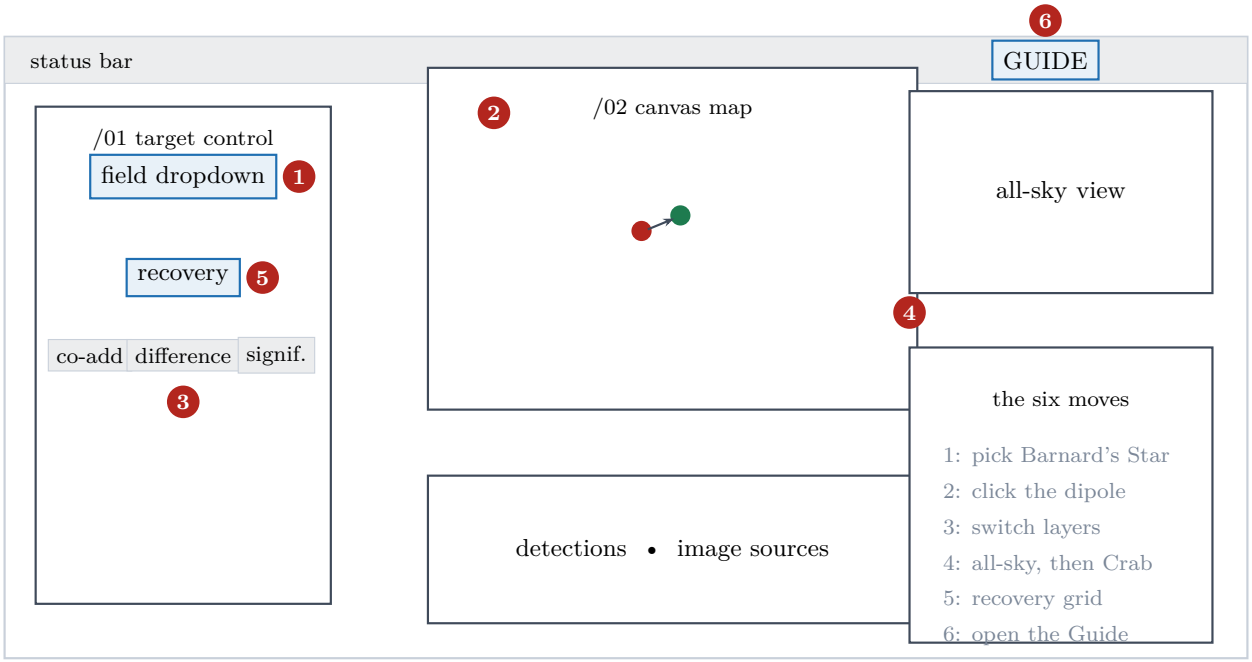
So the result actually being claimed is narrow: seventeen positions where a five-sigma change is unexplained by three catalogues. Not seventeen new objects. The word on the slide is unexplained.

Slide 22 — Live demonstration 1:30

On the slide.

- Open `site/index.html` – already loaded in a second tab, 100% zoom
- **1** pick Barnard's Star, point at the dipole, hover it • **2** click it: verdict, SNR, proper motion
- **3** switch layers: co-add, difference, significance
- **4** all-sky view, then the Crab Nebula (10 detections, all KNOWN)
- **5** recovery grid on the current field • **6** the Guide tab

Figure / diagram. A map of the screen with the six numbered callouts sitting in the gutters beside the exact controls used, so nothing has to be hunted for on stage.



Spoken script. I would like to show this working rather than describe it, so the site is open in the second tab. This is the static version, no server, so everything you are about to see is pre-computed JSON being drawn by the browser.

First, Barnard's Star, which is the demo I would recommend to anyone. The map shows the significance of the difference between two visits. Point at that bright pair in the middle – a dipole, the star's old position in red, its new one in green – and the tooltip gives the verdict, the significance and the catalogues that match. Click it and the verdict panel opens: this is the fifty-seven-sigma detection, a proper motion around twenty-five arcseconds a year, and no catalogue match, for the epoch reason from slide fourteen.

Now the layer switch: co-add, then difference, then significance – the whole chain, visible one step at a time. Over to the all-sky view, where every field is a thumbnail; I will pick the Crab, which gives ten detections and all ten explained, because that is the contrast I want you to see. Finally the recovery grid for the field, which is the completeness matrix, and the Guide tab, which is what a visitor sees before anyone explains anything to them.

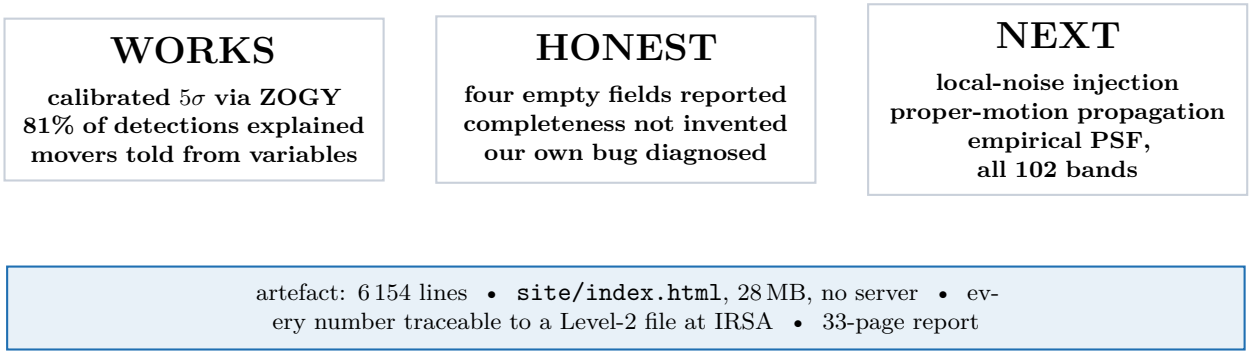
If the laptop misbehaves, the fallback is the same content in the report – figures three, six and seven – and I will move on rather than debug in front of you.

Slide 23 — Conclusion, contributions, and questions 0:50

On the slide.

- **Works**: ZOGY on masked co-adds gives a calibrated 5σ map, morphology separates movers from variables, triage explains 81% of detections
- **Honest**: four fields return nothing, completeness is unmeasured, one candidate is our own bug
- **Next**: local-noise injection, proper-motion propagation, an empirical PSF, all 102 bands
- Artefact: six stages, the derived JSON, a static site, and a guide for non-specialists

Figure / diagram. Three pillars – works, honest, next – with the artefact listed underneath and a compact reference strip at the bottom.



Spoken script. To close, three statements, and then questions.

It works: proper subtraction on masked co-adds gives a significance map whose threshold means something, the morphology separates movers from variables wherever the displacement exceeds the PSF, and catalogue triage explains four fifths of what the pipeline finds.

It is honest: four fields came back empty and the deck says so, the completeness is still unmeasured rather than invented, and the single most significant detection in the sample is a bug found in my own triage and described rather than hidden.

And the next three changes are small, and each one converts a caveat into a number: inject at the image level with local noise, propagate catalogue proper motions before matching, and use an empirical PSF per detector. Everything is in the repository – the six stages, the derived JSON, the static site, and a guide written for someone who has never seen an astronomical image. Thank you; I am happy to take questions.

Backup: five questions and the pre-flight checklist

Questions you are most likely to be asked.

"Why only two or three bands, and not all 102?" Cost: a hundred and two cutouts per epoch per field is roughly 500 MB and seventy minutes of archive time, *per field*. Subtraction does not need the extra bands; the spectral indices do, and that is the first item on the next-steps list.

"Is thirty arcseconds not far too generous a match radius?" For astrometry, yes. For vetoing it is deliberate: a loose radius biases towards KNOWN, which is the conservative direction. The alternative inflates the candidate list and flatters the result.

"What is the real false-positive rate?" Predicted 0.018 pixels per frame at 5σ; the observed detections are tens of sigma above that floor, so the map is not noise-dominated. No measured rate is quoted, because completeness is unmeasured.

"Could this scale to the whole sky?" The CPU could: six seconds per field times 1.8 million nine-arcminute fields is about 125 days on one core. The archive is the bottleneck, which is why the design caches aggressively and pre-renders.

"Why ZOGY rather than SFFT?" ZOGY is closed form and its null distribution is analytic, which is what makes the false-positive budget explicit – the argument on slide twelve. SFFT would be the better choice for an undersampled space PSF, and is listed as future work.

Pre-flight checklist, five minutes before.

- Two browser tabs: `site/index.html` (static, offline-safe) and, if the network is good, the local API on `localhost:8000` for the live fetch.
- Zoom 100%, one field already loaded – Barnard's Star – so the demo does not start on a spinner.
- A screen recording of the demo on the desktop, as the fallback file.
- This document open at page 23, the demo map, so the callouts match the screen.
- Numbers on the tip of the tongue: 18 fields, 128 cutouts, 88 detections, 71 explained, 17 unexplained, 4 empty fields, 0.018 false pixels per frame.